

GitOps / Enterprise Governance

GitOps keeps every site's SparkBridge configuration in a Git repository. Host shows which sites have drifted from it, signs and applies a change to one site or all, rolls back by commit, and records every apply.

Part of HostIgnition 8.1.19+ and 8.3Runs on: Host; sites need Edge with a trusted configuration key

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1. Overview

A bundle is a directory in your repository: `bundle.properties` (version, author, approvals) and one document per site at `sites/<group>/<edge>.properties`. Host reads the checked-out bundle, compares each site's desired version with the version that site declares on its own BIRTH, and reports the difference as drift.

When you apply, Host checks every document, signs each one for its own site with the fleet signing key at a new per-site sequence number, and publishes it on that site's configuration topic. The Edge verifies the signature and the sequence, hot-reloads, and declares the new version on its next BIRTH, which closes the drift.

Review happens where your team already reviews changes: a pull request in your Git host. GitOps enforces an approvals count before it will apply, and records the commit of every apply.

2. What You Need

- A SparkBridge Host gateway (GitOps is built into Host).
- Edge on each site, with `sparkbridge.config.trustedkey` set to the fleet public key. A site without it ignores configuration and stays drifted.
- The fleet signing key on Host at `sparkbridge.host.enrollment.keyfile` (the same key enrollment uses).
- A Git checkout of your bundle on the Host machine, kept current by your own pull (a scheduled `git pull` or your CI).
- A GitOps license to apply. The drift report works on every Host without it.

3. Install

Nothing extra to install: install or upgrade SparkBridge Host from the download page. Clone your configuration repository onto the Host machine, for example to `<ignition>/data/gitops-bundle`.

4. License

GitOps is licensed per Host gateway. Your key names `com.greenguardusa.gitops` and includes Host, SparkCalc and SparkID, so it can be the gateway's one `sparkbridge-license.key` file. A gateway reads one key file: if the same gateway runs another licensed SparkBridge product, ask us for one key that names all of them, because a key that leaves a product out stops that product once its trial has run out. Place the key in the gateway data directory; it is picked up without a restart.

Without the license, `/gitops`, `/gitops/page`, `/gitops/audit` and the drift gauge work as normal. Apply answers 403 with the license it needs and writes a REFUSED audit row, and `mode=apply` runs as a report. While the Host module's Ignition trial is running, apply works so you can evaluate it.

5. Configure

Set these Java system properties on the Host gateway (Ignition `ignition.conf`, one `wrapper.java.additional` line each) and restart once:

<code>sparkbridge.host.gitops.bundleDir</code>	The bundle directory in your checkout. Empty turns GitOps off.
<code>sparkbridge.host.gitops.mode</code>	report (default): show drift only. apply: once a minute, apply to every drifted site that has birthed.
<code>sparkbridge.host.gitops.approvalsRequired</code>	Minimum approvals= in <code>bundle.properties</code> before apply is allowed. Default 1, never below 1.
<code>sparkbridge.host.enrollment.keyfile</code>	The fleet signing key made by <code>ConfigKeys --generate</code> (default <code>data/sparkbridge-config-signing.key</code>). It must be a regular, owner-only file (<code>chmod 600</code>); a key readable by others is refused.

A site document holds the same keys as that site's SparkBridge configuration, and it replaces that configuration as a whole: a key the document leaves out is removed from the site. Start each document from the site's current `sparkbridge-site.properties`, keeping the enrollment keys (assigned name, certificates, username). Set `sparkbridge.config.version` in a document to give it its own version; otherwise it takes the bundle version. Secrets never travel in a bundle: a document with a secret-bearing key is refused, and secrets stay in each site's owner-only secrets file.

6. Operate

1. **Propose.** Change a site document, bump version in `bundle.properties`, and open a pull request.
2. **Review and approve.** Reviewers approve in your Git host; set `approvals=` to the number of approvals it received, then merge.
3. **See drift.** After the checkout on Host pulls the merge, open `/gitops/page`: each site shows desired, applied and why it differs.
4. **Canary.** POST `/gitops/apply?site=Austin/Site12&reason=deadband` with the admin token applies to one site. Watch it reconnect and declare the new version.
5. **Roll out.** POST `/gitops/apply?site=all`, or a comma list of sites. In `mode=apply` Host does this for you once a minute.
6. **Roll back.** Revert or check out the earlier commit in the configured directory and apply again. Host re-signs the earlier documents at a newer sequence, so every site accepts them.

Apply only ever uses the configured bundle directory; it never accepts a different path on the request.

7. Verify

- GET /gitops reports drifted: 0 and each site's applied version equals desired.
- GET /gitops/audit shows one APPLIED row per apply, with the commit, sites, time, reason and the published site@version#sequence.
- The Prometheus gauge sparkbridge_gitops_drift is 0 and sparkbridge_gitops_bundle_error is 0.

8. Troubleshoot

Drift stays open after apply	The site has no sparkbridge.config.trustedkey, or it is not the fleet public key. The site ignores the document.
Applied shows unknown	The site has not sent a BIRTH to this Host since it started. Apply skips unknown sites in mode=apply.
403 "approvals N below required M"	Raise approvals= in bundle.properties after review, or lower approvalsRequired.
403 naming com.greenguardusa.gitops	No GitOps license on this Host. Install the key; no restart needed.
409 bundle refused	A document failed its check (unknown value, secret key, bad path, symlink). The message names the site and key. One bad document refuses the whole bundle.
"document changed under unchanged version"	A document was edited without bumping its version. Bump it; sites ignore a same-version resend.
Known issue: drift after an apply that changes no live setting	An Edge re-announces its configuration version to the Host only when a live setting changes value. An apply that changes no live setting (the site already has those values, or only settings that take effect after a restart changed) is accepted and saved by the Edge, but the Host shows the site as drifted until the Edge reconnects or restarts. A fix is planned for 3.1.1.
"no fleet signing key"	Set sparkbridge.host.enrollment.keyfile to a readable key.

9. Reference

GET /gitops	Drift report as JSON (read token), including the bundle's Git commit and whether apply is licensed.
GET /gitops/page	The same report as a page.
POST /gitops/apply?site=<group/edge,... all> [&reason=]	Apply (admin token, GitOps license).
GET /gitops/audit[?limit=n]	Audit rows, newest last, up to 1,000. File: data/sparkbridge-gitops-audit.jsonl, owner-only.
Sequence file	data/sparkbridge-gitops.seq, shared with enrollment and the PublishConfig tool. Point PublishConfig --sequence-file at it when you run it on the Host machine.
Command line	java -cp common.jar io.sparkbridge.gitops.SparkGitOps build --bundle <dir> --key <file> [--out <dir>] [--publish [--broker <uri>]] [--approvals-required n] builds and optionally publishes the same signed documents off the gateway.

The audit records the credential tier that made the call (for example `rest-admin`) and the bundle author; the named person is the commit author in your Git history.

10. Support

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